

- 5 (a) Explain why the electric potential near an isolated proton is positive.

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..... [3]

- (b) An isolated metal sphere is positively charged and has radius R , as shown in Fig. 5.1.

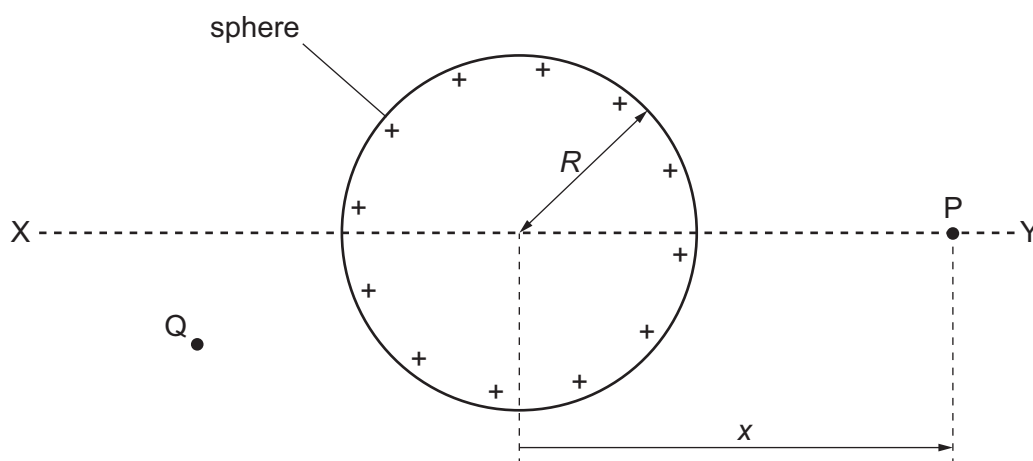


Fig. 5.1

Line XY passes through the centre of the sphere.

Point P lies on line XY at a variable displacement x from the centre of the sphere.

Point Q is at a fixed position that is not on line XY.

The electric field strength at the surface of the sphere is E_0 .

- (i) On Fig. 5.1, draw an arrow at point Q to show the direction of the electric field at that point. [1]

